

HAMMAD ZAHID

AI Engineer | Data Analyst | Machine Learning & Generative AI

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PROFESSIONAL SUMMARY

Entry-level AI engineer and data analyst with hands-on experience building machine learning, data analytics, generative AI, computer vision, and predictive analytics applications using Python. Experienced in transforming raw datasets into analytical insights, interactive dashboards, predictive models, and user-facing AI applications. Currently pursuing a BS in Software Engineering while working as a part-time Computer Science teacher for grades 9-12, which sharpens the ability to break down complex technical ideas into clear, usable explanations. Strong interest in AI engineering, generative AI, multi-agent systems, and building end-to-end ML products rather than notebooks alone. Seeking an entry-level role or internship in Machine Learning, AI Engineering, or Data Analysis.

CORE SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Jupyter, Data Cleaning, EDA, Feature Engineering, Statistical Analysis

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Random Forest, Logistic Regression, Classification, Regression, Clustering, Cross-Validation, Hyperparameter Optimization, Time-Series Forecasting

Generative AI & Agents: LLMs, RAG, LangChain, LangGraph, Prompt Engineering, Ollama, Embeddings, Vector Search, FAISS, AI Agents, Tool Calling, Document QA

Computer Vision & Deep Learning: TensorFlow, Keras, PyTorch, YOLOv8, Faster R-CNN, RetinaNet, CNN, Transfer Learning, OpenCV, Image Classification, Object Detection

Applications & Deployment: Streamlit, Flask, FastAPI, REST APIs, SQLAlchemy, Docker, Git, GitHub, Google Colab, Kaggle

Business & Visualization: Excel, Power BI, Tableau, Interactive Dashboards, KPI Analysis, Business Reporting, Data Storytelling

PROFESSIONAL EXPERIENCE

Data Analyst — Decodelabs

March 2026 – April 2026 | Lahore, Pakistan

- Analyzed and cleaned datasets to support business and product decision-making, applying structured data-quality checks and profiling.
- Built visualizations and summary reports to communicate trends and findings to non-technical stakeholders.
- Worked with Python-based tooling (Pandas, NumPy) to prepare and validate data for downstream analysis.

Data Science Intern — Pakalign Studio

September 2025 – March 2026 | Lahore, Pakistan

- Contributed to data science and applied AI project work, including data preparation, exploratory analysis, and model experimentation.
- Collaborated with team members on technical deliverables spanning data pipelines and analytical reporting.
- Applied Python, SQL, and machine learning tooling to real-world datasets over a multi-month engagement.

Data Analyst — Elevvo Pathways

October 2025 – December 2025 | Remote

- Titanic Data Analysis: cleaned missing values (titles, family size) and visualized survival patterns by class and gender.
- Hearing Survey Analysis: performed data preprocessing and exploratory analysis to uncover trends in hearing health data.
- LinkedIn Web Scraping: built a scraper to collect data science job postings and analyzed in-demand skills, salaries, and market opportunities.
- Retail Sales Analysis: conducted EDA on a retail dataset, identifying product demand patterns and sales trends.

SELECTED PROJECTS

AI Data Analyst Pro — Automated Data Analytics Platform

Python | Flask | SQLAlchemy | Machine Learning | LLMs | AutoML | Forecasting

- Built an end-to-end AI-powered analytics platform transforming raw tabular data into automated analysis, visualizations, predictions, and business reports.
- Implemented multi-format ingestion (CSV, Excel, JSON, SQL, TXT) and automated data-quality workflows covering missing values, duplicates, and schema inference.
- Integrated conversational AI for natural-language dataset analysis; added ML workflows (Random Forest, XGBoost, LightGBM, ARIMA, Prophet) with Docker-based deployment.

GitHub: <https://github.com/Hamad-Ansari/-AI-Data-Analyst-Pro-flask-app>

DataInsight Pro — Automated EDA & Analytics Platform

Python | Pandas | Plotly | Scikit-learn | Streamlit

- Developed an interactive EDA platform for CSV, Excel, and JSON datasets with automated profiling, correlation analysis, and outlier detection.
- Added time-series diagnostics (trend, seasonality, stationarity, ACF/PACF) and ML-oriented diagnostics including feature importance and clustering.

GitHub: <https://github.com/Hamad-Ansari/-DataInsight-Pro>

Customer Churn Prediction

Python | XGBoost | LightGBM | CatBoost | Feature Engineering

- Developed an end-to-end churn prediction workflow on structured customer/telecom data, engineering financial, usage, and tenure-risk features.
- Evaluated XGBoost, LightGBM, and CatBoost via stratified cross-validation, reporting an AUC of approximately 0.9157; combined models via ensembling.

GitHub: <https://github.com/Hamad-Ansari/-Predict-Customer-Churn---Kaggle-Grandmaster-Solution>

Network Intrusion Detection System

Python | Scikit-learn | XGBoost | Streamlit | Cybersecurity

- Built an ML-based intrusion detection app on the KDD Cup 1999 dataset with seven classifiers (Random Forest, Decision Tree, Gradient Boosting, AdaBoost, HistGradientBoosting, KNN, XGBoost).
- Added CSV batch and manual prediction interfaces with confidence scores and feature-importance visualization in an interactive Streamlit dashboard.

GitHub: <https://github.com/Hamad-Ansari/-Network-Intrusion-Detection-System-NIDS->

Object Detection with YOLOv8, Faster R-CNN & RetinaNet

Python | PyTorch | Detectron2 | YOLOv8 | OpenCV

- Built a computer-vision pipeline comparing YOLOv8, Faster R-CNN, and RetinaNet on Roboflow datasets (YOLO/COCO formats) using PyTorch and Detectron2.
- Evaluated performance with mAP@50, mAP@50-95, precision-recall curves, and confusion matrices using Colab GPU environments.

GitHub: <https://github.com/Hamad-Ansari/-Object-Detection-with-YOLOv8-Faster-R-CNN-RetinaNet->

CNN-Based Pneumonia Detection on Chest X-Rays

Python | TensorFlow | Keras | CNN

- Built a convolutional neural network to classify chest X-ray images for pneumonia detection, applying image preprocessing and augmentation.
- Evaluated model performance using accuracy, precision, recall, and confusion matrices.

Brain Tumor Detection using Deep Learning

Python | TensorFlow/PyTorch | CNN | Transfer Learning

- Developed a deep-learning image classification pipeline to detect brain tumors from MRI scans using CNN and transfer learning approaches.

AI vs. Human Text Detection

Python | Scikit-learn | NLP

- Built an NLP/ML classifier to distinguish AI-generated text from human-written text, covering feature extraction and model evaluation.

QanoonGPT — Offline AI Legal Assistant

Python | Streamlit | Ollama | RAG | FAISS | Sentence Transformers

- Built a bilingual Urdu/English AI assistant for retrieving information from Pakistani legal documents using local LLM inference (Ollama) for privacy-focused offline use.
- Developed embeddings-based document retrieval with FAISS and citation-oriented legal question answering.

GitHub: <https://github.com/Hamad-Ansari/-Pakistan-Legal-AI-Assistant>

Professional Stock Forecasting Dashboard

Python | Streamlit | Prophet | Scikit-learn | Plotly | Yahoo Finance

- Built an interactive stock-market analytics dashboard with candlestick visualizations and Prophet/Random Forest-based forecasting workflows.

GitHub: <https://github.com/Hamad-Ansari/Professional-Stock-Forecasting-Dashboard>

Paddy Disease Classification

Python | TensorFlow | Keras | EfficientNetV2-B3

- Developed a deep-learning image classification workflow to identify paddy leaf diseases using EfficientNetV2-B3 with transfer learning.

GitHub: <https://github.com/Hamad-Ansari/Paddy-Disease-Classification-using-Deep-Learning->

CERTIFICATIONS

- Artificial Intelligence on Microsoft Azure — Microsoft (August 2025)
- Foundations of Data Science — Google (August 2025)
- Work Smarter with Microsoft PowerPoint — Microsoft (September 2025)
- What is Data Science?
- Statistical Analysis of Data for Researchers
- Bank of America — Global Markets Sales and Trading Analyst Job Simulation
- Python ka Chilla for Data Science
- Tableau Certified

EDUCATION

Bachelor of Science, Software Engineering

Virtual University of Pakistan | 2023 – 2027

TARGET ROLES

AI Engineer • Machine Learning Engineer • AI/ML Intern • Data Analyst • Junior Data Scientist • Generative AI Engineer • ML Engineer Intern • Business/Data Analyst • AI Engineering Intern